

AE-419: Radiation of water with liquid phase O2 measurment, analog to AE-418

Date: 2025-01-17

Tags: Radiation O2 PNP

[Ru(CO)(OH)2(PNP)] AE liquidphase

Category: Two-photon

Status: Done

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Reaction scheme/sample structure

H2O

Literature/reference experiments

Literature	https://doi.org/10.1039/D1EE01053K
Reproduction	/
Related experiments	Two-photon - AE-402: Radiation of [Ru(CO)(OH)2(PNP)] in water with liquid phase O2 measurment HTE - AE-414: Preparation of stock solutions for the irradiation of [Ru(bpy)3]Cl2 * 6 H2O

Reagents

Name	CAS/Experiment number	Amount [μmol]	Equivalents	Mass _{theo} [mg]	Mass _{exp} [mg]	Molar mass [g/mol]	Volume [ml]
Milli-Q water	GD-018: Redegassing of H2O	/	/	/	/	/	0.65

Irradiation Parameters

Power measured using [\[Power Meter\] 843-R-USB + 919P-020-12](#) unless specified otherwise.

	Light Source Name	Wavelength [nm]	Power Setting [mW]
First light source	Light Source - LCS-0365-76-22	365 nm	300
Second light source	Light Source - LCS-6500-65-22	white	200

Used beam combiner [Name or None]	Beam Combiner - LCS-BC25-0409
Irradiation distance [cm]	10.5
Thermostat temperature [°C]	25

Stirring speed [rpm]	1000
Irradiation start [s]	2700
Irradiation stop [s]	4140

Procedure/observations

Date	Time	Step	Observations
17.01	9:42	To a 5 mL screw cap schlenk flask water was added	
		The setup from Two-photon - AE-418: Radiation of [Ru(CO)(OH)₂(PNP)] in water with liquid phase O₂ measurment was used without modifications	
	9:43	The Robust oxygen probe was attached to the flask using a bola fitting (diameter = 3 mm) according to Protocol - O₂ Measurement procedure	
	9:44	The flask was moved to the setup	
	9:46	The logging was started (AE-419-1-Ch2)	AE-419-1-Ch2.png
	9:46	The flask was flushed with argon for 180 s (strong Ar flow in gas phase)	when the PTFE tubing from the Ar line was connected (after 35 s) the O ₂ value increased quickly to 290 µM from 260 µL, when after approx. 15 s (total time after 50 s) the cannula was punched through the septum the value dropped fast, due to the start of the flushing c(O ₂) dropped during the flushing to approx. 100 µM
	9:50	The Schlenk was degassed according to Protocol - Degassing and O₂ Measurement with a sensor cap with air-stable substances in the Photochem. setup	c(O ₂) dropped quickly to approx. 0 µM. Afterwards slow decrease, in general exponential decay c(O ₂) at the end -0.17 µM
	10:19	The flask was flushed with argon for 240 s (strong Ar flow in gas phase)	sharp increase till 0.9 µM O ₂ in 5 to 10 s, afterwards slower

	10:23	The degassing stopped by removing the PTFE cannula according to Protocol - Degassing and O2 Measurement with sensor cap with air stable substances	increase continued a bit, max. c(O2) approx. 1.1 μM
	10:24	The Log (-1-Ch2) was stopped	AE-419-1-Ch2.txt
	10:46	The logging was started (AE-418-2-Ch2)	
	10:46 - 11:11	The sample was allowed to equilibrate according to Protocol - Degassing and O2 Measurement with sensor cap with air stable substances	Equilibration duration: 2700 s slow decrease in c(O2) from 1.1 to 1.0 over 45 min
	11:11	The light sources were turned on	Sharp decrease (10 s) to 0 μM instant after turning on the light, than slow exponential decrease till -0.08 μM in approx. 1.5 min Than slow increase til end of irradiation -0.03 μM .
	11:35	The light source was turned off	Irradiation duration: 1440 s Instant, sharp increase in c(O2) in 10 s to 0.3 in 1 min to 0.5
		The sample was allowed to equilibrate according to Protocol - Degassing and O2 Measurement with sensor cap with air stable substances	Equilibration duration: 1262 s After 10 min c(O2) = 0.8 μM as max. value, afterwards slow decrease, analog to decrease of pre-reaction baseline
		The logging was stopped (-3-Ch2)	AE-419-2-Ch2.txt AE-419-2-Ch2.png

Analysis

Date	Time	Sample name	Analysis method	Analytic device	Solvent	Raw Data	Processed Data	Comparative Data	Interpretation
17.01	9:46	AE-419-1-Ch2	Optical O2 detection	Equipment - Firesting Fiber-Optic Oxygen Meter 2 Channel	H2O	AE-419-1-Ch2.txt	AE-419-1-Ch2.png	Two-photon - AE-418: Radiation of [Ru(CO)(OH)2(PNP)] in water with liquid phase O2 measurment (AE-418-1-Ch2)	For detailed interpretation see Observations in Procedures/Observation table above
17.01	10:26	AE-419-2-Ch2	Optical O2 detection	Equipment - Firesting Fiber-Optic Oxygen Meter 2 Channel	H2O	AE-419-2-Ch2.txt	AE-419-2-Ch2.png	Two-photon - AE-418: Radiation of [Ru(CO)(OH)2(PNP)] in water with liquid phase O2 measurment (AE-418-2-Ch2)	For detailed interpretation see Observations in Procedures/Observation table above

Result

Very similar behaviour as for AE-418, therefor "wired" behaviour are artefacts, e.g. the decrease upon irradiation and the increase at the end

Also increase when stopping degassing observed --> likely artefact

Linked experiments

- AE-236: Irradiation of PhPDA (AE-234), 1.5 mg/mg SDS, 2 mg/mL PhPDA
- AE-237: Irradiation of PhPDA (AE-234), 1.5 mg/mg SDS, 2 mg/mL PhPDA I
- AE-245: Synthesis of poly(1,4-diphenylbutadiyne)
- AE-249: Irradiation of PhPDA (AE-245), 1.5 mg/mg SDS, 2 mg/mL PhPDA
- AE-254: Irradiation of PhPDA (AE-245), 1.5 mg/mg SDS, 2 mg/mL PhPDA reproduction
- AE-257: Synthesis of poly(1,4-diphenylbutadiyne)
- AE-258: Irradiation of PhPDA (AE-257), 1.5 mg/mg SDS, 2 mg/mL PhPDA
- AE-259: Irradiation of PhPDA (AE-257), 1.5 mg/mg SDS, 2 mg/mL PhPDA after 1 d
- AE-260: Irradiation of PhPDA (AE-257), 1.5 mg/mg SDS, 2 mg/mL PhPDA after 2 d
- AE-262: Irradiation of PhPDA (AE-257), 1.5 mg/mg SDS, 2 mg/mL PhPDA after 3 d
- AE-276: Synthesis of poly(1,4-diphenylbutadiyne), 1.2 g scale
- AE-279: Irradiation of PhPDA (AE-276-2), 1.5 mg/mg SDS, 2 mg/mL PhPDA
- AE-282: Irradiation of PhPDA (AE-276-1), 1.5 mg/mg SDS, 2 mg/mL PhPDA
- GD-018: Redegassing of H₂O
- AE-313: Synthesis of PPy via Fenton reaction with FeSO₄ and 10 eq. H₂O₂ I
- AE-332: Irradiation of PPy (AE-312), 2 mg/mL
- AE-333: Irradiation of PPy (AE-313), 2 mg/mL
- C3N₄ - AE-398: Irradiation of C3N₄-Co-0.2 from Lingli Ni
- C3N₄ - AE-399: Irradiation of C3N₄-Co-0.4 from Lingli Ni I

HTE - AE-414: Preparation of stock solutions for the irradiation of $[\text{Ru}(\text{bpy})_3]\text{Cl}_2 \cdot 6 \text{H}_2\text{O}$

Two-photon - AE-358: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-310) 365 nm and white, 50 mM KOH

Two-photon - AE-382: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ using Ag_2O , 5 h

Two-photon - AE-386: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-382) 365 nm and white, 50 mM KOH in D_2O I

Two-photon - AE-402: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP})]$ in water with liquid phase O_2 measurement

Two-photon - AE-418: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP})]$ in water with liquid phase O_2 measurement

Linked resources

Archived - [Degassing procedure for photocatalytic water splitting by polymeric photocatalysts](#)

Beam Combiner - [Beam Combiner LCS-BC25-0409 -1](#)

Beam Combiner - [Beam Combiner LCS-BC25-0409 -2](#)

Equipment - [Firesting Fiber-Optic Oxygen Meter 2 Channel \(Firesting 1\)](#)

Equipment - [Freezer Lab E004 CEEC II](#)

Light Source - [UHP LED 405 nm](#)

Light Source - [UHP LED 6500 K \(white\)-1](#)

Light Source - [HP LED 405 -2](#)

Light Source - [HP LED 6500 K \(white\) -2](#)

Protocol - [Irradiation setup](#)

Protocol - [O2 Measurement procedure](#)

Protocol - [UV vis measurements \(BioTek PowerWave HT\)](#)

Protocol - [Determination of H2O2 by Spectroquant](#)

Protocol - [Degassing and O2 Measurement with sensor cap with air stable substances](#)

Protocol - [Determination of H2O2 in the suspensions of photocatalysts using H2O2 test strips](#)

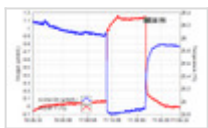
Protocol - [Ultrasonication with the UP 200st from hilscher in ZAF 208](#)

Protocol - [H2O2 photometric detection with DPD-POD method for irradiated suspensions of photocatalysts](#)

Attached files

AE-419-2-Ch2.png

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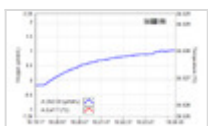


AE-419-2-Ch2.txt

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AE-419-1-Ch2.png

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AE-419-1-Ch2.txt

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Unique eLabID: 20250117-cfc84f769c518495401bf45c4555d1da80249f90

Link: <https://elab.water-splitting.org/experiments.php?mode=view&id=1644>